

Real-World Dual-Robot Platforms

Robot 0 (Unitree Go2 quadruped) and Robot 1 (wheeled chassis) share one indoor scene; each carries its own RGB-D/LiDAR sensing and onboard compute.

ROBOT 0 — Unitree Go2 (quadruped)

RealSense D435i

RGB-D, head-mounted, forward-facing

Livox Mid-360

360° spinning LiDAR, top-mounted

Odin1 spatial memory module

RGB-D camera + LiDAR + IMU · Manifold Tech
manifoldtech.cn

ROBOT 1 — wheeled chassis

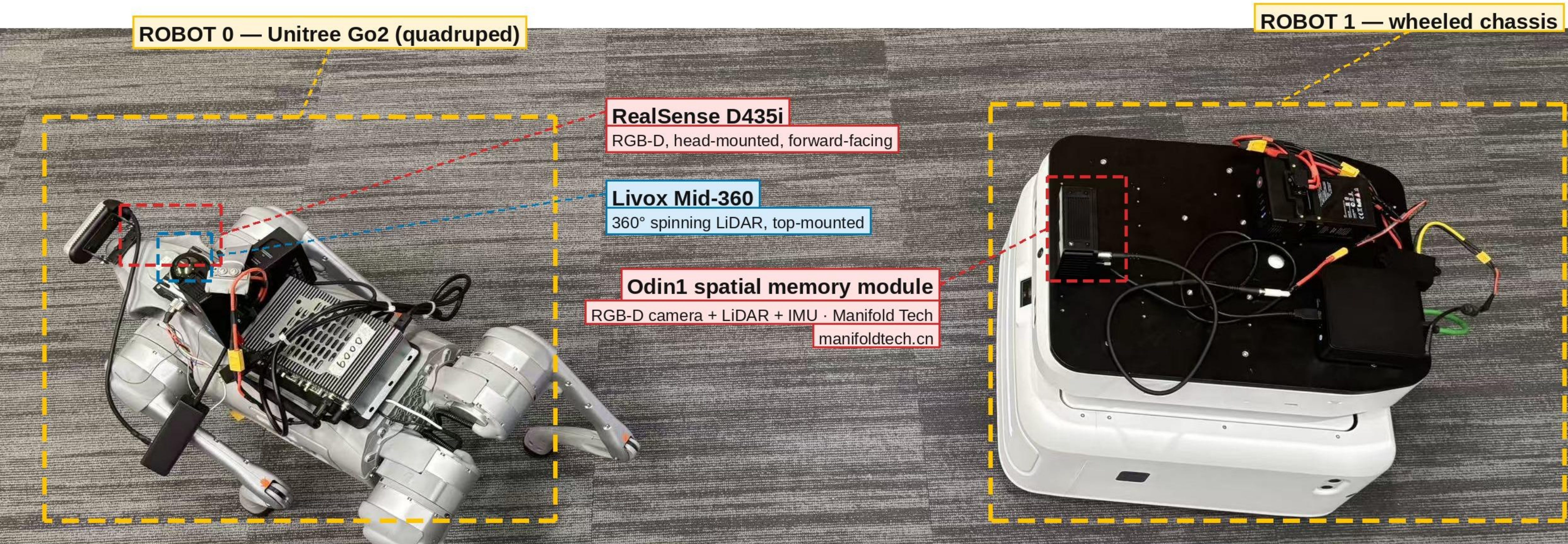


Fig. Real-world experiment platforms. Robot 0 is a Unitree Go2 quadruped carrying a head-mounted Intel RealSense D435i RGB-D camera, a top-mounted Livox Mid-360 LiDAR, and onboard compute. Robot 1 is a wheeled delivery chassis carrying an Odin1 spatial memory module (RGB-D camera + LiDAR + IMU, Manifold Tech, manifoldtech.cn) and onboard compute on its rear deck. Both platforms run local planning/control onboard; the hub only publishes targets.

Shared Cross-Robot Calibration

Both robots observe one shared circle-grid board from different positions to register their local frames into one shared frame.

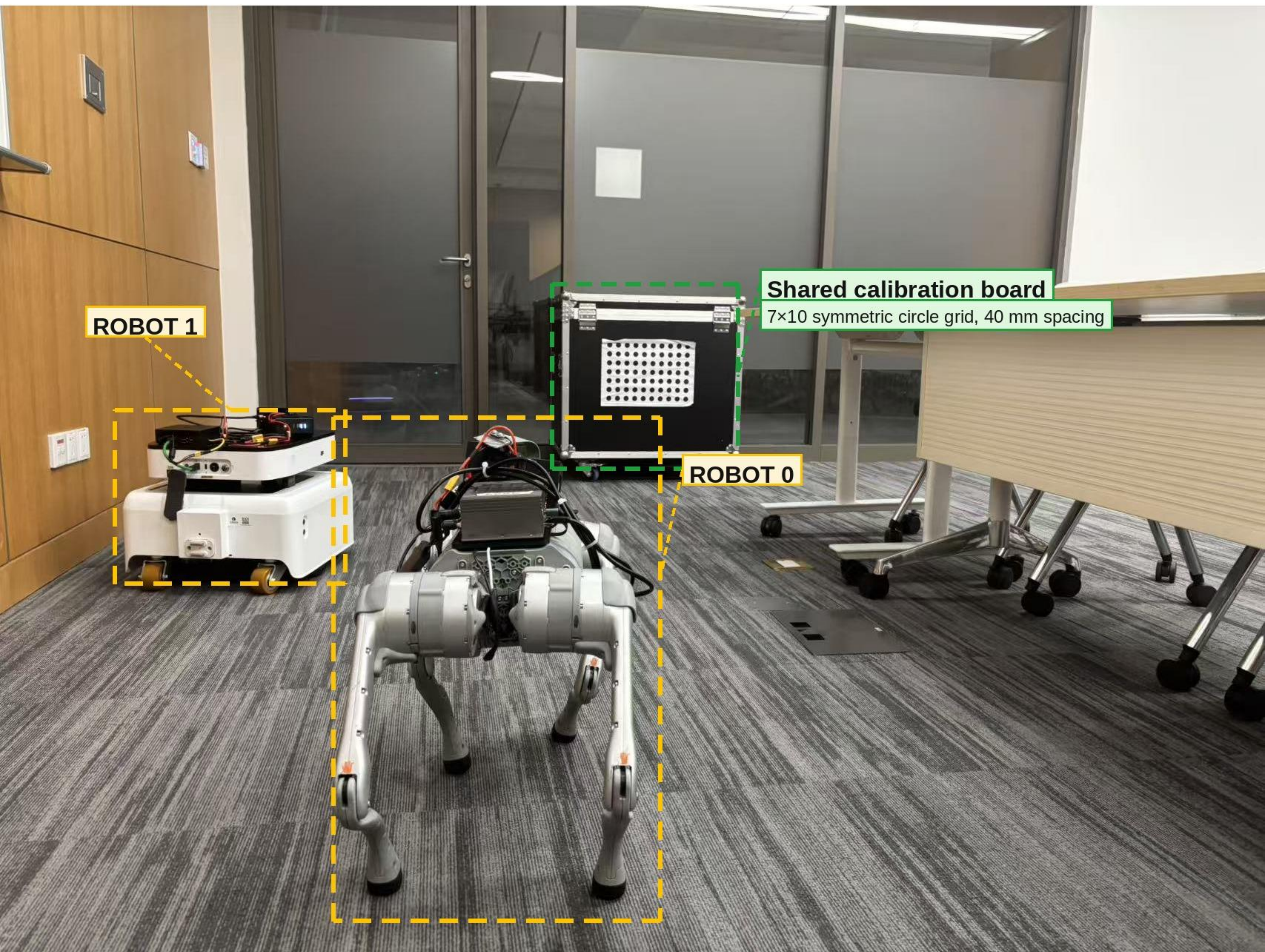


Fig. Cross-robot shared-frame calibration. With both robots stationary, each robot's own camera captures the shared board and solves its pose by PnP; the two per-camera poses compose into a fixed transform between the robots' camera frames. The board is then moved at least 10 cm or rotated at least 5° and re-observed as an independent holdout — the calibration is only accepted if it also explains this second, independently moved observation.